

XUANYU TIAN

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Curriculum Vitae (Jun. 2026)

RESEARCH INTERESTS

Inverse Problems in Medical Imaging

Accelerated MRI, Motion Correction, CT Reconstruction, Dynamic Imaging, *etc.*

Representation Learning, Generative Modeling, Optimization

Implicit Neural Representation, Diffusion Models, Plug-and-Play Prior *etc.*

EDUCATION

ShanghaiTech University

Shanghai, China

Ph.D. Candidate in Computer Science

2021 – Exp. 12/2026

- Advisor: Prof. Yuyao Zhang

Wuhan University of Technology

Wuhan, China

B.E. in Computer Science and Technology

2017 – 2021

WORKING EXPERIENCE

AI Healthcare Lab, Ant Group | Hangzhou, China

5/2026 – Now

Research Intern

- Host: Dr. Le Lu

AWARDS AND HONORS

China National Scholarship ($\approx \$4,300$)

12/2025

Ministry of Education

Outstanding Student (Top 5%)

12/2024

ShanghaiTech University

Outstanding Student (Top 5%)

12/2023

ShanghaiTech University

PUBLICATIONS

Conference Publications (“*” indicates equal contribution)

15. C. Du, X. Tian, Q. Wu, M. Liu, J. Yu, H. Wei, and Y. Zhang, “Plug-and-play diffusion meets ADMM: Dual-variable coupling for robust medical image reconstruction,” in *Proc. 43rd Int. Conf. Mach. Learn. (ICML)*, Seoul, South Korea, Jul. 2026.
★ Poster [Acceptance Rate: $\approx 26.6\%$]
14. M. Liu, X. Tian, C. Du, Q. Wu, H. Wei, and Y. Zhang, “Resolving blind inverse problems under dynamic range compression via structured forward operator modeling,” in *Proc. 43rd Int. Conf. Mach. Learn. (ICML)*, Seoul, South Korea, Jul. 2026.
★ Poster [Acceptance Rate: $\approx 26.6\%$]
13. H. Zhang, Q. Wu, X. Tian, B. Li, Y. Zhang, and H. Wei, “Reference-free meta-learning for generalized implicit neural representation in efficient MRI reconstruction,” in *Proc. 43rd Int. Conf. Mach. Learn. (ICML)*, Seoul, South Korea, Jul. 2026.
★ Poster [Acceptance Rate: $\approx 26.6\%$]
12. C. Du, Q. Wu, X. Tian, J. Yu, Y. Zhang, and H. Wei, “Improving 2D diffusion models for 3D medical imaging with inter-slice consistent stochasticity,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Rio de Janeiro, Brazil, Apr. 2026.
★ Poster [Acceptance Rate: $\approx 28\%$]
11. X. Wang, J. Wu, X. Tian, T. Sun, H. Wei, and Y. Zhang, “Spatio-temporal continuous fetal body atlas via latent-conditioned implicit neural representation,” in *Proc. IEEE Int. Symp. Biomed. Imag. (ISBI)*, London, UK, Apr. 2026, pp. 1–5.
10. J. Yang*, Y. Cao*, X. Tian, Y. Zhang, and H. Wei, “NICE: Neural implicit craniofacial model for orthognathic surgery prediction,” in *Proc. AAAI Conf. Artif. Intell. (AAAI)*,

Singapore, Jan. 2026.

★ Poster [Acceptance Rate: $4167/23680 = 17.6\%$]

9. X. Tian, L. Chen, Q. Wu, X. Wang, J. Feng, Y. Zhang, and H. Wei, “Unsupervised motion-compensated decomposition for cardiac MRI reconstruction via neural representation,” in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, Singapore, Jan. 2026.
★ Poster [Acceptance Rate: $4167/23680 = 17.6\%$]
8. S. Cui, Y. Lian, X. Tian, Y. Huang, J. Yu, Y. Zhang, and X. Cai, “Full-wave inversion for speed-of-sound imaging in ultrasound tomography via self-supervised learning,” in *Proc. IEEE Int. Ultrason. Symp. (IUS)*, Utrecht, Netherlands, Sept. 2025.
7. Q. Wu*, C. Du*, X. Tian, J. Yu, Y. Zhang, and H. Wei, “Moner: Motion correction in undersampled radial MRI with unsupervised neural representation,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Singapore, Apr. 2025.
★ Spotlight [Acceptance Rate: $587/11500 = 5.1\%$]
6. X. Li, X. Tian, X. Zhao, H. Wei, and Y. Zhang, “Accelerated 3D thermometry field reconstruction from tri-planar images via cross-view implicit representation,” in *Proc. IEEE Int. Symp. Biomed. Imag. (ISBI)*, Houston, TX, USA, Apr. 2025.
★ Oral Presentation
5. X. Tian, L. Chen, Q. Wu, C. Du, J. Shi, H. Wei, and Y. Zhang, “Unsupervised self-prior embedding implicit neural representation for iterative sparse-view CT reconstruction,” in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, Philadelphia, PA, USA, Feb. 2025.
★ Poster [Acceptance Rate: $3032/12957 = 23.4\%$]
4. X. Lin*, C. Du*, Q. Wu, X. Tian, J. Yu, Y. Zhang, and H. Wei, “Zero-shot low-field MRI enhancement via denoising diffusion driven neural representation,” in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, Marrakesh, Morocco, Oct. 2024.
3. S. Lai, X. Tian, Q. Wu, C. Du, X. Xu, H. Wei, X. Guan, and Y. Zhang, “Reconstructing knee CT volumes from biplanar X-rays via self-supervised neural field,” in *Proc. IEEE Int. Symp. Biomed. Imag. (ISBI)*, Athens, Greece, May 2024.
2. C. Jiang*, X. Tian*, Y. Li, J. Wu, X. Mu, L. Zhang, and Y. Zhang, “Self-supervised high-dimensional magnetic resonance image denoising using super-resolved single noisy image,” in *Proc. IEEE Int. Symp. Biomed. Imag. (ISBI)*, Cartagena de Indias, Colombia, Apr. 2023.
1. X. Tian, Q. Wu, H. Wei, and Y. Zhang, “Noise2SR: Learning to denoise from super-resolved single noisy fluorescence image,” in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, Singapore, Sept. 2022.

Journal Publications (“*” indicates equal contribution)

4. L. Chen*, X. Tian*, J. Wu, R. Feng, G. Lao, Y. Zhang, H. Liao, and H. Wei, “Joint coil sensitivity and motion correction in parallel MRI with a self-calibrating score-based diffusion model,” *Med. Image Anal. (MedIA)*, vol. 102, Art. no. 103502, 2025.
3. X. Tian, J. Wu, G. Lao, C. Du, C. Jiang, Y. Li, J. Zhang, H. Wei, and Y. Zhang, “Self-supervised denoising for high-dimensional magnetic resonance image,” *Biomed. Signal Process. Control (BSPC)*, vol. 104, Art. no. 107451, 2025.
2. L. Chen*, X. Tian*, J. Wu, G. Lao, Y. Zhang, and H. Wei, “COLLATOR: Consistent spatial-temporal longitudinal atlas construction via implicit neural representation,” *Med. Image Anal. (MedIA)*, vol. 100, Art. no. 103396, 2024.
1. X. Tian, Z. Dong, X. Lin, Y. Gao, H. Wei, Y. Ma, J. Yu, and Y. Zhang, “Zero-shot image denoising for high-resolution electron microscopy,” *IEEE Trans. Comput. Imag. (IEEE TCI)*, vol. 10, pp. 1462–1475, 2024.

Pre-prints (“*” indicates equal contribution)

3. Q. Wu, X. Tian, C. Du, H. Zhang, X. Wang, L. Lu, and Y. Zhang, “Disentangled learning improves implicit neural representations for medical reconstruction.” Pre-print, arXiv:2605.04234, 2026.

2. T. Yu, **X. Tian**, J. Yang, D. He, J. Yu, X. Wang, and Y. Zhang, "SPIDER: Structure-preferential implicit deep network for biplanar X-ray reconstruction." Pre-print, arXiv:2507.04684, 2025.
1. C. Du*, X. Lin*, Q. Wu, **X. Tian**, Y. Su, Z. Luo, H. Wei, S. Zhou, J. Yu, and Y. Zhang, "DPER: Diffusion prior driven neural representation for limited-angle and sparse-view CT reconstruction." Pre-print, arXiv:2404.17890, 2024.

RESEARCH EXPERIENCE

Unsupervised Reconstruction for Cardiac MRI [AAAI'26] 3/2025 – Present

- Proposed an unsupervised reconstruction framework for free-breathing cardiac MRI, to mitigate motion artifacts from arrhythmia and failed breath holds.
- Decomposed dynamic sequences into a static template and time-varying deformation fields, modeled with INRs and optimized using undersampled k -space data.
- Achieved high temporal resolution reconstruction (20-40ms) enabling precise cardiac and respiratory motion modeling validated on multi-vendor in vivo data (Siemens Avanto, UIH uMR 790).

MRI Motion Correction [ICLR'25 (Spotlight)][MedIA'25] 9/2024 – Present

- Physiological motion during long MRI acquisitions often causes severe artifacts, undermining image quality and diagnostic reliability.
- Proposed unsupervised methods that unifies physics-based motion modeling and diffusion priors, enabling motion-free image directly from corrupted k -space data.
- Achieved SOTA performance on multi-sequence (2D/3D) in vivo data from a UIH uMR 790 scanner, markedly enhancing robustness in complex clinical scenarios.

SNR-Robust Reconstruction for Sparse-View CT [AAAI'25] 9/2024 – 12/2025

- Proposed an unsupervised method that integrates INRs with PnP optimization, enabling stable optimization and high-fidelity recovery.
- Outperformed prior unsupervised and supervised methods across diverse dose levels and acquisition settings, with strong cross-condition generalization.

Self-Supervised Image Denoising [TIP'24][BSPC'25][MICCAI'22] 2021 – 2024

- Integrated super-resolution techniques to preserve fine structural details and introduced a random sub-sampling strategy to disrupt spatial noise correlations
- **Representative applications:**
 - 1) Poisson–Gaussian denoising (*e.g.*, fluorescence imaging);
 - 2) High-dimensional medical imaging (*e.g.*, BOLD fMRI, DWI);
 - 3) Ultra-low SNR imaging (*e.g.*, transmission electron microscopy (TEM)).

TEACHING EXPERIENCE

CS270B: Advanced Digital Image Processing Spring 2023 *ShanghaiTech University*

- Position: Teaching Assistant
- Instructor: Prof. Yuyao Zhang

ACADEMIC SERVICES

Journal Reviewer for:

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- Expert Systems with Applications (ESWA)

Conference Reviewer/PCs for:

- ICML / CVPR / ECCV / AAAI
- MICCAI / IEEE ISBI

SKILLS

Languages: Mandarin (native), English (fluent).

Programming: Python, MATLAB, L^AT_EX.

REFERENCES

Prof. Yuyao Zhang

Associate Professor, ShanghaiTech University

Relationship: Advisor

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